

Control as Inference, Inference by Architecture:

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Abstract

Robot learning has long benefited from injecting useful inductive biases into neural models: geometric structure, dynamics, feedback, temporal smoothness, task hierarchies, and other assumptions that make learning more sample efficient, interpretable, and robust. This remains important, but it is no longer sufficient. As robotics moves toward large pre-trained vision–language–action models, many of the most important inductive biases are already present before the roboticist begins: inherited from architectures, simulators, data mixtures, action parameterisations, objectives, optimisation procedures, and deployment software. The central problem is therefore shifting from designing all of the relevant structure ourselves to discovering, characterising, and controlling the structure already embedded in learned robot policies. This paper argues that control theory provides a natural language for this task. A large robot policy can be viewed as a learned feedback system whose internal representations, circuits, manifolds, latent goals, and action distributions determine how information is transformed into behaviour. Probability provides the calculus for representing uncertainty, latent variables, multimodality, demonstrations, and trajectories, while architecture provides the prior, whether that prior was deliberately engineered or inherited from pretraining. We develop this position through behaviour cloning, score-based policy learning, and teacher quality in imitation learning, and argue that observability, excitation, stability, and causal intervention should be treated as core tools for analysing modern robot learning systems. In this view, interpretability is not simply post-hoc visualisation; it is system identification for learned robot controllers.

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1 An apology

This paper is an indulgence. It reflects my own attempts to make sense of a set of ideas that have appeared repeatedly in my work and in the work of collaborators from whom I have learned a great deal. It is not intended to be a comprehensive survey, and it almost certainly misses important related work. Good ideas are common, often rediscovered, and usually improved by many people at once; I will use this as my excuse.

A second apology is also needed. I used generative AI while drafting this document. The thoughts, prejudices, and narrative tics are mostly mine. I think. It is becoming harder to tell.

I should also stress that I am not a control engineer, nor a statistician, nor a machine learning or computer vision expert. I am an academic, so dabble in these things, and try to do the best I can to put the bits and pieces I’ve picked up along the way together.

2 Introduction: the modelling problem in robot learning

Modern robot learning has benefited enormously from the ability to train expressive neural models from data. Policies can now be conditioned on images, force signals, proprioception, language, task context, and other sensory streams, and can produce actions ranging from torques and velocities to pose targets, impedance parameters, subgoals, and symbolic decisions. This flexibility has allowed learned systems to replace or augment parts of the classical perception-planning-control pipeline in a wide range of settings.

However, robotics differs from many other machine learning problems in an important way: robots act in the systems from which their future data is generated. A prediction error can produce a poor action; a poor action can move the robot into an unfamiliar state; the resulting observation can then be further from the training distribution. Small errors can therefore become closed-loop failures. This is not merely a problem of function approximation, but of modelling dynamical systems under feedback.

It is tempting to write a learned policy as

$$\pi_{\theta} : \mathcal{O} \rightarrow \mathcal{U}, \quad (1)$$

where observations are mapped directly to controls. While useful, this notation hides much of the structure that makes robotics difficult. The observation may be an image, a tactile signal, a force trace, a pose estimate, or a history of such quantities. The control may be a low-level actuation command, a desired pose, a compliance parameter, or a high-level action. Between these two spaces lies a sequence of implicit modelling choices: what is observed, what is remembered, what is estimated, what is planned, and what is controlled.

A policy is therefore better viewed as a learned computation graph,

$$\mathcal{O} \xrightarrow{\text{observer}} \mathcal{Z}_1 \xrightarrow{\text{estimator}} \mathcal{Z}_2 \xrightarrow{\text{planner}} \mathcal{Z}_3 \xrightarrow{\text{controller}} \mathcal{U}, \quad (2)$$

where each intermediate representation lives in a space chosen, either explicitly or implicitly, by the modeller. The central question is not whether we should use models or learning. Practical robot systems use both. The more useful question is:

Which assumptions should be built into the model, and which should be learned from data?

This paper argues that control theory and probability give us a natural language for asking this question. Control is concerned with systems that evolve over time, interact with their environments, and transform information into action under feedback [4, 46]. These are also the defining features of deployed robot learning systems. Concepts such as observability, excitation, stability, robustness, controllability, identifiability, and feedback interconnection are therefore not separate from deep learning for robotics; they are names for the failure modes and modelling requirements that appear when learned policies are placed in the loop.

The AI slop position advanced here is:

Robot learning is structured distribution learning under feedback.

However, the emphasis of this paper is not only on designing this structure before training. It is also on identifying the structure that is already present after training, and asking whether it is observable, excited, stable, causal, and safe. This perspective connects a thread running through much of our work: from explicitly designed visual servoing and feedback controllers [16, 17], to latent physical representations [3, 39, 40], to structured demonstration learning [18, 20, 28], and more recently to learned latent feedback controllers and interpretable VLA policies [80] and hierarchical diffusion policies [?].

3 From optimality-variable inference to distributional modelling

A common formulation of *control as inference* introduces binary optimality variables and frames control as posterior inference conditioned on optimality. Under this construction, a trajectory

distribution is weighted by an exponential reward term, and a policy is obtained by inferring actions conditioned on the event that the trajectory is optimal. This perspective gives a coherent probabilistic interpretation of maximum-entropy control, reinforcement learning, and approximate inference [45, 50, 66, 74, 75].

This paper does not argue against that view. It is elegant, useful, and has shaped much of modern robot learning. However, it is not the only way to understand control as inference. A broader interpretation is that any robot learning system defines a probabilistic model, whether or not this model is explicitly written down. It defines distributions over observations, states, actions, trajectories, rewards, teachers, dynamics, parameters, and internal representations. The modelling task is to decide how these distributions factorise, which parts should be specified, which parts should be learned, and which parts should be amortised by a neural architecture.

Let

$$\tau = (x_{0:T}, o_{0:T}, u_{0:T}) \quad (3)$$

be a trajectory of latent states, observations, and actions. A generic robot learning model may be written as

$$p_\theta(\tau) = p(x_0) \prod_{t=0}^T p_\theta(o_t \mid x_t) p_\theta(u_t \mid \mathcal{I}_t) p_\theta(x_{t+1} \mid x_t, u_t), \quad (4)$$

where $\mathcal{I}_t$ denotes the information available to the policy at time t . Different robot learning problems correspond to different commitments about this factorisation. Behaviour cloning focuses on the action distribution,

$$p_\theta(u_t \mid o_{\leq t}), \quad (5)$$

model-based reinforcement learning learns or specifies the dynamics,

$$p_\theta(x_{t+1} \mid x_t, u_t), \quad (6)$$

inverse reinforcement learning attempts to infer a reward or task model,

$$p_\theta(r_t \mid x_t, u_t), \quad (7)$$

and filtering estimates beliefs over latent states,

$$p_\theta(x_t \mid o_{\leq t}, u_{< t}). \quad (8)$$

These are often treated as distinct problem classes. The view taken here is that they are better understood as different factorisations of a common structured distribution. The modelling problem is to choose a family

$$\mathcal{P}_\Theta = \{p_\theta(x_{0:T}, z_{0:T}, u_{0:T}, o_{0:T}) : \theta \in \Theta\} \quad (9)$$

that is expressive enough to represent the behaviours of interest, but constrained enough to be learned from finite data and to remain useful when placed in a feedback loop.

This is the point at which inductive bias becomes unavoidable.

4 Inductive bias as executable prior information

An inductive bias is usually described as an assumption that allows a learning system to generalise. In robotics, this description is a bit too weak. A useful inductive bias should say something about which behaviours, explanations, representations, dynamics, or feedback interconnections are plausible before data is observed.

In probabilistic terms, an inductive bias may be expressed as a prior over parameters,

$$p(\theta), \tag{10}$$

or, more importantly for neural robot learning, as a prior over functions, trajectories, and closed-loop behaviours induced by the architecture,

$$p(\pi_\theta), \quad p_\theta(\tau). \tag{11}$$

This is closely related to Bayesian interpretations of neural networks, where priors over weights induce priors over functions [53, 54]. It is also consistent with model-based machine learning, where modelling assumptions are made explicit and manipulated as part of the design process [7, 78].

The same idea appears throughout modern deep learning. Convolutional networks exploit translation structure. Group-equivariant networks generalise this idea to broader symmetry groups [25]. Graph networks encode relational structure and provide a formal account of relational inductive bias [5]. Geometric deep learning frames many successful architectures as ways of respecting symmetries, locality, scale, and compositionality in the data [11]. Even architectures that are not explicitly probabilistic can behave like priors: the deep image prior shows that network structure alone can favour natural image statistics before task-specific learning [76], and spectral-bias analyses show that neural networks can prefer some functions over others during optimisation [65].

A neural network architecture is therefore not simply an implementation detail. It is a modelling decision. Its layers, bottlenecks, recurrence, equivariances, residual connections, attention mechanisms, differentiable simulators, energy functions, and controller structures define which mappings are easy to represent, which gradients are easy to follow, and which closed-loop behaviours are likely after training.

In robotics, this point has a particularly concrete interpretation. A physics-informed model, a movement primitive, a differentiable simulator, a graph representation, a latent state-space model, a diffusion policy, and a low-level feedback controller all restrict the space of behaviours that the learner can express. Some of these restrictions are deliberately engineered. Others are inherited from pretraining, data mixtures, action parameterisations, optimisation procedures, or deployment infrastructure. In either case, the model is not learning in a vacuum. It is learning inside a structured family of possible explanations and behaviours.

In this sense, inference is performed by architecture. A neural network does not merely approximate an inference procedure after the fact. It defines the family of hypotheses within which inference takes place:

Architecture is prior information made executable.

For robotics, the relevant priors are rarely only smoothness or simplicity. They include priors about geometry, contact, inertia, causality, locality, time, feedback, task structure, embodiment, hierarchy, and stability. A model-based approach to robot learning should therefore not ask whether to use learning or control. It should ask which parts of the control problem should be made explicit in the architecture, objective, data collection process, inference procedure, robot morphology, or deployment stack.

This view also reframes a thread running through much prior robot learning work. Dynamic movement primitives represent skills as learnable nonlinear dynamical systems with attractor structure [38, 68]. Probabilistic movement primitives represent distributions over trajectories rather than single nominal motions [58]. Information-theoretic policy search constrains policy updates by relative entropy, thereby imposing a geometry on the search process itself [60]. Deep Lagrangian Networks encode Lagrangian mechanics into the model class, so that learned robot dynamics inherit physical

structure rather than being represented as unconstrained regressors [52]. Sparse identification, Koopman methods, and dynamic mode decomposition similarly search for representations in which dynamics become identifiable, parsimonious, and useful for control [13–15, 64]. These examples differ technically, but they share a common modelling principle: learning is made useful by restricting the space in which explanations and behaviours are allowed to live.

5 Behaviour cloning as conditional distribution learning

Behaviour cloning provides a useful perspective to expand upon this. Consider a dataset of demonstrations

$$\mathcal{D} = \{(o_i, u_i)\}_{i=1}^N, \quad (12)$$

where observations may include images, touch, velocities, force measurements, pose estimates, proprioception, task context, or any other available sensor signal. Behaviour cloning is often formulated as supervised learning,

$$\theta^* = \arg \min_{\theta} \frac{1}{N} \sum_{i=1}^N \ell(\pi_{\theta}(o_i), u_i). \quad (13)$$

This perspective has been central to learning from demonstration, from early neural driving systems to modern robot policies [2, 63]. In our own work, the limitations of treating demonstrations as simple input–output pairs appear in several forms: demonstrations can encode compositional programs [18], relational structure [35], residual corrections to existing movement primitives [28], or implicit task progress and reward information [20].

However, the supervised learning notation hides an important modelling choice. The demonstrator does not necessarily define a single action for every observation. There may be many good actions, the teacher may be exploratory, and the task may be partially observed. Behaviour cloning is therefore more naturally written as conditional distribution learning,

$$\pi_{\theta}(u \mid o) \approx p_{\text{data}}(u \mid o), \quad (14)$$

with maximum-likelihood objective

$$\theta^* = \arg \min_{\theta} \mathbb{E}_{(o,u) \sim \mathcal{D}} [-\log \pi_{\theta}(u \mid o)]. \quad (15)$$

This distinction matters in robotics. If several valid actions are available, a deterministic mean-squared-error policy may learn their average, which need not itself be valid. If the demonstrator is exploring, early actions may be information gathering rather than reward maximising, which is precisely the setting considered when learning rewards from exploratory demonstrations [20]. If the task is partially observed, the same current observation may require different actions depending on unobserved context. Moreover, the sequential nature of imitation learning means that prediction errors change future observations, producing compounding error under deployment [67].

A distributional policy makes these modelling issues explicit. We must decide whether π_{θ} is Gaussian, mixture-based, energy-based, diffusion-based, autoregressive, hierarchical, or constrained by a controller. Each choice defines a different prior over behaviour.

6 Score matching and policies as action distributions

One way to represent rich action distributions is to learn the score of the conditional density rather than the density itself. In this case, a policy model learns

$$s_\theta(u, o) \approx \nabla_u \log p_{\text{data}}(u \mid o). \quad (16)$$

Score matching provides an objective for fitting unnormalised density models without evaluating the normalising constant [37]. With denoising score matching [77], demonstrated actions are perturbed as

$$\tilde{u} = u + \sigma \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (17)$$

and the score model is trained using

$$\theta^* = \arg \min_{\theta} \mathbb{E}_{(o, u) \sim \mathcal{D}, \epsilon} \left[\left\| s_\theta(\tilde{u}, o, \sigma) + \frac{\tilde{u} - u}{\sigma^2} \right\|_2^2 \right]. \quad (18)$$

This is useful because many robotic behaviours are naturally multimodal. A robot wiping a surface, inserting a tool, manipulating a deformable object, or scanning a patient may have many locally valid action sequences. In such settings, the goal is not to regress to an average action, but to represent a structured distribution over plausible actions and then sample, optimise, or condition this distribution under feedback. Modern score-based and diffusion models provide one mechanism for doing this [23, 34, 70]. This should be viewed as part of a broader pattern: mixture models, switching models, residual models, and feedback-structured policies all provide different ways of representing multimodal or hybrid robot behaviour [19, 28, 80].

This changes the interpretation of a policy. A policy is not merely a command generator. It is a local behavioural model,

$$\pi_\theta(u_t \mid \mathcal{I}_t), \quad (19)$$

which specifies what actions are plausible given the information available at time t . When deployed, this conditional distribution is coupled to the environment dynamics, producing a closed-loop trajectory distribution.

7 Neural policies as modular inference systems

A neural policy can be written as a composition of functions,

$$\pi_\theta(o) = f_{\theta_L}^{(L)} \circ f_{\theta_{L-1}}^{(L-1)} \circ \dots \circ f_{\theta_1}^{(1)}(o). \quad (20)$$

Equivalently,

$$z_0 = o, \quad z_\ell = f_{\theta_\ell}^{(\ell)}(z_{\ell-1}), \quad z_\ell \in \mathcal{Z}_\ell. \quad (21)$$

The policy is therefore a cascade of representation maps. In robotics, these maps often correspond to familiar components,

$$\pi_\theta = c_{\theta_c} \circ p_{\theta_p} \circ h_{\theta_h} \circ b_{\theta_b} \circ e_{\theta_e}, \quad (22)$$

where

$$e_{\theta_e} : \mathcal{O} \rightarrow \mathcal{X} \quad \text{observer or encoder,} \quad (23)$$

$$b_{\theta_b} : \mathcal{X} \rightarrow \mathcal{S} \quad \text{state estimator or belief updater,} \quad (24)$$

$$h_{\theta_h} : \mathcal{S} \rightarrow \mathcal{G} \quad \text{goal generator or high-level planner,} \quad (25)$$

$$p_{\theta_p} : \mathcal{S} \times \mathcal{G} \rightarrow \mathcal{A} \quad \text{action or motion planner,} \quad (26)$$

$$c_{\theta_c} : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{U} \quad \text{low-level controller.} \quad (27)$$

A stochastic version of the same architecture may be written as

$$p_{\theta}(u, \hat{s}, g, a \mid o) = p_{\theta_e}(u \mid \hat{s}, a)p_{\theta_p}(a \mid \hat{s}, g)p_{\theta_h}(g \mid \hat{s})p_{\theta_b}(\hat{s} \mid e_{\theta_e}(o)). \quad (28)$$

This factorisation is a modelling assumption. It says that the policy should solve the task by observing, estimating, selecting goals, planning actions, and controlling actuation. The modules may be neural, differentiable, probabilistic, hand-designed, learned, or hybrid. The important point is that the architecture expresses a hypothesis about how the control problem decomposes.

This is not an argument for returning to brittle hand-engineered pipelines. It is an argument for treating modularity as a modelling choice. Earlier visual servoing and tracked-vehicle systems made the feedback structure explicit [16, 17]; latent physical state spaces can make proportional controllers useful from pixels [40]; residual policies can combine movement primitives with learned corrections for contact-rich tasks [28]; surgical cutting models can use task-specific physical constraints and movement primitives to structure demonstrations [71, 72]; and probabilistic reward models can infer task structure from exploratory demonstrations [20]. In each case, the architecture constrains the distribution that is learned.

8 A taxonomy of inductive biases for robot learning

The distributional view suggests a taxonomy of inductive biases according to which part of the robot learning model they constrain. The point of such a taxonomy is not to draw hard boundaries. In practice, the same modelling choice often affects perception, dynamics, action, data collection, and stability at once. The purpose is instead to ask where prior information enters the learning system, and how it shapes the closed-loop behaviour that eventually emerges.

8.1 Observation and representation biases

Observation and representation biases constrain how raw sensor data becomes a state, belief, feature, or latent variable,

$$z_t = \phi_{\theta}(o_{\leq t}). \quad (29)$$

Examples include convolutional encoders, equivariant networks, object-centric representations, graph networks, keypoint bottlenecks, physical latent spaces, temporal filters, causal feature selection, and multimodal sensor fusion. Their purpose is to improve observability by making task-relevant state easier to infer.

This connects robot learning to a broader literature on representation structure. Group-equivariant networks build symmetry into perception [25]. Graph networks encode relational structure between entities [5]. Geometric deep learning treats invariance and equivariance as organising principles for architecture design [11]. In manipulation, spatial inductive biases have been especially productive. Transporter Networks formulate manipulation as spatial displacement in visual feature space, while equivariant variants explicitly encode rotational structure so that manipulation knowledge can transfer across object orientations [36, 79].

In our own work, physical latent variables inferred from video, video-based system identification, Newtonian latent spaces, and visual structure discovery for system identification use related assumptions: the observation model should not merely compress pixels, but infer variables that behave like physical states and therefore support prediction, goal identification, and control [3, 39–41].

8.2 Dynamics biases

Dynamics biases constrain the transition model,

$$p_{\theta}(x_{t+1} \mid x_t, u_t). \quad (30)$$

Examples include structured state-space models, differentiable simulators, residual dynamics, Hamiltonian or Lagrangian networks, contact models, hybrid switching systems, and Koopman representations. Their purpose is to improve data efficiency and extrapolation by modelling what is already known about the physics.

There is a long tradition of data-driven system identification that is directly relevant here. Sparse identification of nonlinear dynamics discovers parsimonious governing equations from trajectory data, and its controlled variants extend this idea to input-output systems [14, 15]. Dynamic mode decomposition with control and Koopman-based approaches search for representations in which nonlinear systems admit useful low-dimensional or approximately linear descriptions for analysis and control [13, 64]. These methods are not usually framed as robot policy learning, but they are philosophically close to the present argument: useful learning depends on finding representations in which the relevant dynamics become identifiable, controllable, and interpretable.

Neural dynamics models provide a complementary lineage. Neural ordinary differential equations parameterise continuous-time dynamics directly [22]. Hamiltonian Neural Networks and Lagrangian Neural Networks encode physical structure such as conservation laws or variational mechanics into the model class [27, 30]. Deep Lagrangian Networks apply this idea to robot dynamics by learning models that preserve Lagrangian structure [52]. Vid2Param, physics-as-inverse-graphics, and vision-based system identification follow the same broad principle: use data, including visual data, but constrain the explanation through a dynamical model [3, 39, 41].

8.3 Policy and controller biases

Policy and controller biases constrain the action distribution,

$$\pi_{\theta}(u_t \mid \mathcal{I}_t). \quad (31)$$

Examples include impedance controllers, proportional controllers, motion primitives, mixture policies, diffusion policies, hierarchical policies, safety filters, residual controllers, and latent feedback controllers. Their purpose is to ensure that actions remain plausible and robust when the learned model is placed in feedback with the world.

A substantial precursor to modern structured policies is the movement-primitives literature. Dynamic movement primitives represent skills as nonlinear dynamical systems with attractor structure, making stability and modulation part of the policy representation itself [38, 68]. Policy search for motor primitives shows that the choice of policy parameterisation determines which behaviours are easy to discover [48, 49]. Probabilistic movement primitives represent distributions over trajectories, allowing variability, conditioning, and blending to be handled probabilistically [58]. Relative entropy policy search constrains policy updates by information geometry, preventing learning from making uncontrolled jumps in policy space [60]. These approaches differ from modern deep policies, but they make the same point: the robot does not search over all possible behaviours, but over behaviours expressible through a structured motor representation.

Modern generative policies extend this idea to richer action distributions. Implicit behavioural cloning uses energy-based action models to avoid collapsing multimodal demonstrations to a mean action [29]. Behaviour Transformers represent multimodal behaviour using discrete-continuous action structure [69]. Diffusion policies learn denoising processes over action sequences, providing expressive

conditional distributions over plausible robot behaviour [23]. Switching density networks, residual learning from demonstration, and switching latent feedback controllers similarly use controller structure to represent hybrid or contact-rich skills [19, 28, 80].

8.4 Reward and task biases

Reward and task biases constrain the task model,

$$p_{\theta}(r_t \mid x_t, u_t, c_t). \quad (32)$$

Examples include learned reward functions, inverse optimal control, preference models, temporal ranking, success classifiers, symbolic task constraints, and language-conditioned objectives. Their purpose is to connect weak or ambiguous task specifications to actionable control objectives.

This class of bias is important because tasks are often not directly observable in the robot’s sensor stream. A human demonstrator may know why an action is being taken, but the robot may only observe what was done. Inverse reinforcement learning, preference learning, and imitation learning all address this gap in different ways. DAgger corrects the distribution-shift problem caused by supervised imitation under closed-loop deployment [67]. Generative adversarial imitation learning reframes imitation as matching occupancy measures rather than regressing to individual actions [33]. Broader algorithmic treatments of imitation learning emphasise that the data, teacher, objective, and policy class jointly define what can be learned from demonstrations [57]. Probabilistic temporal ranking and mutual-information-based reward learning are examples of using weak, exploratory, or expert demonstrations to infer task structure [20, 44].

8.5 Data and excitation biases

Data and excitation biases constrain the data collection process,

$$\mathcal{D} \sim d^{\beta}(o, u). \quad (33)$$

Examples include active learning, exploration bonuses, uncertainty sampling, teacher-quality estimation, causal interventions, coverage metrics, curriculum design, and embodiment-aware data collection. Their purpose is to ensure that the data excites the behaviours the learned policy must reproduce or improve upon.

This is the point at which system identification and robot learning meet most directly. Parameters cannot be identified from data that does not excite the relevant directions of the system. Likewise, a policy cannot be expected to behave reliably in closed-loop regimes that were never visited during data collection. In large-scale robot learning this issue is partly hidden by dataset size, but not removed. The data mixture, embodiment distribution, simulator, teleoperation interface, task curriculum, and action representation all determine which behaviours are visible to the model and which are absent.

8.6 Stability and robustness biases

Stability and robustness biases constrain the closed-loop system,

$$\xi_{t+1} = \mathcal{F}_{\theta}(\xi_t, w_t). \quad (34)$$

Examples include Lyapunov constraints, contraction metrics, passivity-inspired layers, bounded-gain networks, robust model-predictive-control wrappers, control barrier functions, and uncertainty-aware action selection. Their purpose is to prevent small errors from becoming catastrophic failures.

There is a growing literature that reconnects learned policies with formal notions of safety and stability. Neural Lyapunov methods learn controllers and Lyapunov functions while using falsification to search for counterexamples [21]. Control barrier functions provide optimisation-based safety filters through forward-invariance constraints [1]. Safe learning methods use uncertainty-aware models, Gaussian processes, and Lyapunov arguments to expand the region of attraction without leaving verified safe sets [6, 12]. These approaches are complementary to the present argument: they show how learned components can be embedded in feedback systems whose safety properties are explicitly tested rather than merely hoped for.

The recent latent-feedback-controller view provides a concrete version of the same idea. Learned behaviours are safer when representations are actively driven toward stable latent goals rather than allowed to drift away from demonstrated behaviour [80].

8.7 Embodiment and morphology biases

Finally, some inductive biases are not implemented in software at all. The robot body itself constrains the learning problem. Compliance, damping, kinematic structure, actuator bandwidth, sensor placement, contact geometry, and morphology determine which behaviours are easy, stable, or even observable.

This connects to older arguments about embodiment and morphological computation: the body is not merely the plant to be controlled, but part of the computation [32, 61, 62]. A compliant mechanism can simplify control. A poor sensor layout can make a task unobservable. A useful end-effector can turn a difficult manipulation problem into an easy one. In this sense, morphology is also an inductive bias, but one implemented in hardware rather than in a loss function or neural architecture.

9 Variational inference and amortised control structure

The previous taxonomy described inductive biases according to the part of the robot learning model they constrain. Another way to describe the same modelling choices is through inference. If a robot learning system contains latent states, goals, modes, rewards, dynamics parameters, or internal controller variables, then deployment requires some form of posterior inference:

$$p_{\theta}(z_{0:T} \mid o_{0:T}, u_{0:T}), \quad (35)$$

or, more generally,

$$p_{\theta}(z_{0:T}, g_{0:T}, m_{0:T}, x_{0:T} \mid o_{0:T}, u_{0:T}, c_{0:T}). \quad (36)$$

For most useful models this posterior is not available in closed form. The usual response is to introduce an approximate posterior,

$$q_{\phi}(z_{0:T} \mid o_{0:T}, u_{0:T}), \quad (37)$$

and to learn it jointly with the generative model.

This is the standard technique in variational inference. Given observations y , latent variables z , and a generative model $p_{\theta}(y, z) = p_{\theta}(y \mid z)p_{\theta}(z)$, the marginal likelihood is

$$\log p_{\theta}(y) = \log \int p_{\theta}(y, z) dz. \quad (38)$$

Since this integral is usually intractable, we introduce $q_\phi(z \mid y)$ and optimise the evidence lower bound,

$$\log p_\theta(y) \geq \mathbb{E}_{q_\phi(z|y)} [\log p_\theta(y, z) - \log q_\phi(z \mid y)] \quad (39)$$

$$= \mathbb{E}_{q_\phi(z|y)} [\log p_\theta(y \mid z)] - D_{\text{KL}}(q_\phi(z \mid y) \parallel p_\theta(z)). \quad (40)$$

The first term asks the latent variable to explain the data. The second term prevents the approximate posterior from drifting too far from the prior. The prior may be simple, such as a standard Gaussian, or structured, such as a dynamical prior, a controller prior, a physical prior, or a switching prior.

A common simplification is the mean-field approximation,

$$q_\phi(z \mid y) = \prod_{k=1}^K q_{\phi_k}(z_k \mid y), \quad (41)$$

which assumes that the approximate posterior factorises across latent variables. This is often computationally convenient, but it is also a modelling assumption. It says that posterior dependencies between latent variables are either weak, irrelevant, or too expensive to represent. In robotics this assumption is rarely innocent. Position, velocity, contact mode, object pose, task phase, and controller state are usually coupled. A mean-field posterior may therefore make inference easy while destroying precisely the structure needed for control.

The alternative is not necessarily to abandon approximation, but to choose approximations that respect the structure of the robot learning problem. For sequential data, a more appropriate factorisation is often

$$q_\phi(z_{1:T} \mid o_{1:T}, u_{1:T}) = \prod_{t=1}^T q_\phi(z_t \mid z_{<t}, o_{\leq t}, u_{<t}), \quad (42)$$

or, when smoothing is available,

$$q_\phi(z_{1:T} \mid o_{1:T}, u_{1:T}) = \prod_{t=1}^T q_\phi(z_t \mid z_{<t}, o_{1:T}, u_{1:T}). \quad (43)$$

These factorisations turn inference into a learned filtering or smoothing problem.

This is where amortisation enters. Rather than solving a new optimisation problem for every trajectory, we learn an inference network,

$$q_\phi(z \mid y) = \text{encoder}_\phi(y), \quad (44)$$

which maps observations directly to an approximate posterior. The cost of inference is paid during training, and deployment uses a fast learned approximation. In robotics language, the encoder is not merely a feature extractor. It is an amortised observer.

This gives another version of the central argument:

Amortised inference is learned state estimation under an architectural prior.

To see this in a toy sequential setting, consider a latent-variable model for observations and actions,

$$z_t \sim p_\theta(z_t \mid z_{t-1}, u_{t-1}), \quad (45)$$

$$o_t \sim p_\theta(o_t \mid z_t), \quad (46)$$

$$u_t \sim p_\theta(u_t \mid z_t). \quad (47)$$

The latent variable z_t may represent a physical state, a belief state, a task phase, or a controller state. The inference problem is to estimate

$$p_\theta(z_t \mid o_{\leq t}, u_{< t}), \quad (48)$$

which is exactly the form of a filtering problem. A variational approximation replaces this with

$$q_\phi(z_t \mid o_{\leq t}, u_{< t}), \quad (49)$$

implemented by a recurrent encoder, temporal convolution, transformer, or structured state-space model.

A variational recurrent neural network makes this explicit [24]. In one simplified form,

$$h_t = f_\theta(h_{t-1}, z_{t-1}, o_{t-1}, u_{t-1}), \quad (50)$$

$$p_\theta(z_t \mid h_t) = \mathcal{N}(\mu_t^p, \Sigma_t^p), \quad (51)$$

$$q_\phi(z_t \mid h_t, o_t, u_t) = \mathcal{N}(\mu_t^q, \Sigma_t^q), \quad (52)$$

$$p_\theta(o_t, u_t \mid z_t, h_t) = p_\theta(o_t \mid z_t, h_t)p_\theta(u_t \mid z_t, h_t). \quad (53)$$

The corresponding sequential ELBO is

$$\mathcal{L}(\theta, \phi) = \sum_{t=1}^T \mathbb{E}_{q_\phi(z_t \mid h_t, o_t, u_t)} [\log p_\theta(o_t, u_t \mid z_t, h_t)] - D_{\text{KL}}(q_\phi(z_t \mid h_t, o_t, u_t) \parallel p_\theta(z_t \mid h_t)). \quad (54)$$

The recurrent state h_t stores history. The prior $p_\theta(z_t \mid h_t)$ predicts what latent state should come next. The approximate posterior $q_\phi(z_t \mid h_t, o_t, u_t)$ corrects that prediction using the current measurement. This is almost embarrassingly control-theoretic: prediction, innovation, correction, and action generation are all present, but implemented as differentiable neural modules.

NewtonianVAE provides a more physically committed toy example [40]. Instead of allowing the latent state to be arbitrary, it biases the representation toward second-order physical structure. A simplified version writes

$$z_t = \begin{bmatrix} q_t \\ \dot{q}_t \end{bmatrix}, \quad (55)$$

with latent dynamics resembling

$$\dot{q}_{t+1} = \dot{q}_t + \Delta t a_\theta(q_t, \dot{q}_t, u_t), \quad (56)$$

$$q_{t+1} = q_t + \Delta t \dot{q}_{t+1}. \quad (57)$$

The decoder maps latent physical states back to pixels,

$$p_\theta(o_t \mid z_t), \quad (58)$$

while the encoder provides an approximate posterior,

$$q_\phi(z_t \mid o_t). \quad (59)$$

The ELBO has the same general form as before,

$$\mathcal{L}(\theta, \phi) = \sum_{t=1}^T \mathbb{E}_{q_\phi(z_t \mid o_t)} [\log p_\theta(o_t \mid z_t)] - D_{\text{KL}}(q_\phi(z_t \mid o_t) \parallel p_\theta(z_t \mid z_{t-1}, u_{t-1})), \quad (60)$$

but the important difference is architectural. The prior is not simply a Gaussian regulariser. It is a statement that useful latent variables should behave like physical states. This makes the learned representation more compatible with feedback control, because proportional or goal-directed control in the latent space becomes meaningful.

This is the broader lesson. Variational inference gives us a training mechanism for latent-variable models. Amortisation gives us fast learned inference. Mean-field and structured variational families specify which posterior dependencies are ignored and which are preserved. Architectures such as VRNNs, NewtonianVAEs, switching latent controllers, and diffusion policies then instantiate different assumptions about what kind of hidden structure is useful for prediction and control.

For robot learning, the relevant question is therefore not just whether the model has latent variables. It is:

What does the variational family allow the robot to infer, and what does the architecture force it to forget?

A poor variational approximation can erase contact modes, task phase, uncertainty, or multimodality. A useful one can turn raw observations into beliefs, physical states, latent goals, or controller variables. In this sense, variational inference is not a side issue in robot learning. It is one of the mechanisms by which architecture becomes prior information, and by which prior information becomes executable behaviour.

10 Control as the language of deep robot learning

Once a neural policy is viewed as part of a closed-loop system, familiar control questions become learning questions. Suppose the world evolves according to

$$x_{t+1} = F(x_t, u_t, w_t), \quad o_t = H(x_t, v_t), \quad u_t = \pi_\theta(o_{\leq t}). \quad (61)$$

If the policy has internal memory or learned representations, we may write

$$z_{t+1} = \Phi_\theta(z_t, o_t), \quad u_t = \Psi_\theta(z_t, o_t). \quad (62)$$

Deployment is then governed by the augmented closed-loop dynamics

$$x_{t+1} = F(x_t, u_t, w_t), \quad (63)$$

$$z_{t+1} = \Phi_\theta(z_t, o_t), \quad (64)$$

$$u_t = \Psi_\theta(z_t, o_t), \quad (65)$$

$$o_t = H(x_t, v_t). \quad (66)$$

The learned model is therefore not only a predictor. It is a feedback interconnection between the physical world and a learned dynamical system. This makes observability, excitation, and stability central to robot learning.

10.1 Observability

Observability asks whether the information stream available to the policy contains the latent variables needed for control. For a learned representation,

$$z_t = \phi_\theta(o_{\leq t}), \quad I(z_t; x_t) \text{ sufficient for control.} \quad (67)$$

A representation cannot recover what the sensor stream does not reveal. In deep learning terms, this is a question of representation sufficiency. In control terms, it is observability. This is why physical latent-space models are useful examples: when latent variables are constrained to behave like states, they can support prediction, goal identification, and control rather than merely compressing observations [3, 39, 40].

10.2 Teacher quality as observability alignment

Learning from demonstration introduces a particular observability problem. The human teacher and the robot may not have access to the same information. A human may use visual cues, tactile sensations, proprioceptive knowledge, semantic context, task intent, or prior experience that is unavailable to the robot at deployment. A demonstration may therefore be expert, but still be a poor training signal for the robot.

Let

$$\mathcal{I}_t^R = \{o_{\leq t}^R, u_{< t}^R\} \quad (68)$$

be the information available to the robot, and let

$$\mathcal{I}_t^H = \{o_{\leq t}^H, u_{< t}^H, c_{\leq t}\} \quad (69)$$

be the information available to the human demonstrator. The human produces actions according to

$$u_t^H \sim p_H(u_t \mid \mathcal{I}_t^H), \quad (70)$$

but the robot must learn a policy of the form

$$\pi_\theta(u_t \mid \mathcal{I}_t^R). \quad (71)$$

A demonstration is well aligned when the human action is explainable from robot-accessible information,

$$u_t^H \perp\!\!\!\perp \mathcal{I}_t^H \mid \mathcal{I}_t^R. \quad (72)$$

Equivalently,

$$I(u_t^H; \mathcal{I}_t^H \mid \mathcal{I}_t^R) \approx 0. \quad (73)$$

Conditional mutual information measures residual dependence after conditioning [26]. If this quantity is large, then the human action is likely to depend on information unavailable to the robot. The demonstration is not wrong, but it may be difficult or impossible for the robot to imitate from its own sensors.

Define the human-only information as

$$\mathcal{I}_t^{H \setminus R} = \mathcal{I}_t^H \setminus \mathcal{I}_t^R. \quad (74)$$

This suggests a teacher-quality weight,

$$w_t = \exp \left(-\lambda I(u_t^H; \mathcal{I}_t^{H \setminus R} \mid \mathcal{I}_t^R) \right), \quad (75)$$

and a weighted behaviour cloning objective,

$$\theta^* = \arg \min_{\theta} \mathbb{E}_{(o_t^R, u_t^H) \sim \mathcal{D}} \left[-w_t \log \pi_\theta(u_t^H \mid \mathcal{I}_t^R) \right]. \quad (76)$$

This is a concrete example of the broader argument. A control concept, observability, becomes a route to improve imitation learning. It suggests when demonstrations are informative, when

they are unlearnable from the robot’s sensors, and how data might be filtered, reweighted, or recollected. This is particularly relevant in expert-demonstration settings, such as robotic ultrasound, where task knowledge is often implicit and demonstrations may be scarce, noisy, exploratory, or expert-specific [20, 44]. It also connects to work on social cue analysis and human-robot interaction, where information-theoretic measures such as transfer entropy can be used to ask which signals actually influence behaviour [42, 43].

A causal version of the same idea asks whether interventions on human-only information would change the demonstrated action. Using the do-operator notation of causal modelling [59], define

$$\text{ACE}(\mathcal{I}_t^{H \setminus R} \rightarrow u_t^H) = \mathbb{E}[u_t^H \mid \text{do}(\mathcal{I}_t^{H \setminus R} = a)] - \mathbb{E}[u_t^H \mid \text{do}(\mathcal{I}_t^{H \setminus R} = b)]. \quad (77)$$

A large causal effect indicates that the teacher is relying on privileged information. A small causal effect indicates that the demonstration is aligned with the robot’s sensing and memory.

10.3 Excitation

Excitation asks whether the data probes the directions of the system that must be learned. For regressors φ_t , a classical persistent excitation condition is

$$\sum_{t=1}^T \varphi_t \varphi_t^\top \succeq \alpha I, \quad \alpha > 0. \quad (78)$$

This condition is central in system identification because parameters cannot be identified from data that does not excite the corresponding directions [51].

For neural policies, an analogous condition is that the dataset excites the directions in parameter or representation space that the model must use,

$$\mathbb{E}_{\mathcal{D}} [\nabla_{\theta} f_{\theta} \nabla_{\theta} f_{\theta}^\top] \succeq \alpha I. \quad (79)$$

A policy cannot be expected to behave reliably in closed-loop regimes that were never excited during data collection.

10.4 Stability

Stability asks whether errors remain bounded when the learned policy is deployed in feedback. For the augmented state

$$\xi_t = \begin{bmatrix} x_t \\ z_t \end{bmatrix}, \quad (80)$$

a Lyapunov-style robust stability condition is

$$V(\xi_{t+1}) - V(\xi_t) \leq -\alpha \|\xi_t\|^2 + \gamma \|w_t\|^2. \quad (81)$$

This is a control-theoretic way of asking whether perception errors, model errors, actuation noise, or distribution shift are attenuated or amplified by the learned feedback loop [4, 46]. It also connects directly to recent work on latent feedback controllers, where stability is studied by analysing whether learned representations are driven toward latent goals or allowed to drift away from demonstrated behaviour [80].

A compact summary is

Observability: $z_t = \phi_{\theta}(o_{\leq t})$, $I(z_t; x_t)$ sufficient, Excitation: $\mathbb{E}_{\mathcal{D}}[\nabla_{\theta} f_{\theta} \nabla_{\theta} f_{\theta}^\top] \succeq \alpha I$, Stability: $V(\xi_{t+1}) - V(\xi_t) \leq -\alpha \ \xi_t\ ^2 + \gamma \ w_t\ ^2$.	(82)
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11 Are engineered inductive biases enough?

The preceding sections defined robot learning as an inference problem under feedback. Control as inference is not only inference over optimality variables. It is the construction and analysis of structured distributions for systems that observe, infer, plan, and act in the world. An architecture is not only a computational implementation of these distributions. It is executable prior information: a set of structural preferences that shapes what behaviours are easy to learn, which representations are likely to emerge, and which feedback patterns are favoured.

Much of model-based robot learning has been motivated by a simple and useful idea: if we know something about the robot, the task, or the world, we should build this knowledge into the model. Geometry, dynamics, contact, temporal structure, feedback, causality, task hierarchy, and embodiment are all useful assumptions. When made explicit, they can improve sample efficiency, robustness, interpretability, and generalisation.

This argument still matters, but it is no longer the whole problem. The next generation of robot learning systems will not always be designed, trained, and inspected from first principles by the roboticist. Increasingly, robots will be controlled by large pre-trained vision-language-action models, trained on mixtures of real robot data, simulated trajectories, human demonstrations, reinforcement learning, internet-scale vision-language data, and proprietary engineering decisions. In these systems, the relevant inductive biases are already present before the downstream roboticist begins. They are inherited from the model architecture, the training data, the simulator, the optimiser, the reward or imitation objective, the action representation, the tokenisation scheme, the sensor interface, the embodiment distribution, and the deployment stack.

This issue is no longer hypothetical. Recent robot foundation models such as RT-1, RT-2, RT-X, OpenVLA, π_0 , and GR00T N1 are trained from heterogeneous mixtures of robot trajectories, language, vision-language data, simulation, human videos, and embodiment-specific action representations [8–10, 47, 55, 56]. Their behaviour is therefore shaped not only by explicit robotics assumptions, but also by data mixtures, action chunking, tokenisation, pretraining corpora, flow or diffusion parameterisations, embodiment metadata, and implementation constraints. These choices are inductive biases, but they are often inherited rather than deliberately designed by the downstream roboticist.

The scientific problem for *roboticists*¹ is therefore shifting:

from injecting inductive biases	→	studying the inductive biases already present.
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Importantly, this is not a rejection of inductive bias, but a broadening of the concept. An inductive bias is not only something we deliberately impose when designing a model. It is any structural preference that shapes the distribution of behaviours a learned robot policy can produce. Some biases are engineered. Some are inherited. Some are discovered only after training. Some are useful. Some are dangerous².

This changes the role of control theory. Control is not only a source of structures to build into neural policies. It becomes a diagnostic language for studying structures that have already emerged inside them. We can ask: what variables are being regulated? What information is available? Which internal states behave like estimates? Where are the latent goals or fixed points? Which circuits causally drive action selection? Which perturbations decay, and which grow? Which trajectories remain on the training manifold, and which drift into unsupported regions?

¹These are different to robotics engineers. The trick to spotting them is to check how many bold numbers are in their tables.

²I grew up in South Africa, and understand this well.

Under this view, a large robot policy is not an unstructured black box. It is a learned physical intelligence system with an implicit anatomy. It contains circuits, manifolds, feedback loops, latent goals, switching behaviours, internal control laws, and inherited priors. These structures may not have been named by the model designer, but they still shape behaviour.

This is also not a push for interpretability as it is commonly understood. The manifolds and latent goals need not map neatly to human abstractions. They are interpretable because they exist, because they affect behaviour, and because they can be measured, perturbed, and analysed. Recent work on mechanistic interpretability for vision-language-action models is a useful example of this shift: the goal is not merely to visualise attention maps, but to identify internal representations and interventions that causally affect robot behaviour [31].

I think the aim of model-based robot learning must therefore expand. We should continue to build useful structure into systems when we can. But for the models that will increasingly control real robots, we also need tools to reverse-engineer the structure they already contain, test whether it is safe, and intervene when it is not.

This motivates the central slogan:

Control is the language; probability is the calculus; architecture is the prior;
interpretability is system identification.

If anything this is needed because everyone and their dog is now working on physics-informed something, and I am bored. Also, many have forgotten the bitter lesson [73], as I do too often.

Control theory gives us a language for studying these biases. Observability asks what information the model has made available to itself. Excitation asks which behaviours, directions, and regimes were sufficiently probed during training. Stability asks whether the learned feedback interconnection attenuates or amplifies errors. Causal analysis asks which internal circuits actually drive actions. Manifold analysis asks where the model’s representations move during execution, and when they leave the support of familiar behaviour.

In this view, interpretability is not simply a post-hoc visualisation problem. It is system identification for learned robot controllers. This suggests a practical research agenda:

1. **Identify the implicit variables.** Determine which internal representations behave like state estimates, goals, controller states, task variables, or contact modes.
2. **Characterise the learned feedback laws.** Ask what is being regulated, where the fixed points are, and how perturbations propagate through the policy.
3. **Map the representation manifolds.** Track where internal trajectories move during execution, and detect when they drift away from the support of training behaviour.
4. **Estimate observability and excitation.** Determine whether the model has access to the information required for control, and whether the data sufficiently probed the behaviours it must perform.
5. **Intervene and verify.** Use causal interventions, steering, projection, safety filters, controller wrappers, or retraining procedures to test and modify the structures discovered inside learned policies.

12 Challenges and open questions

12.1 Formal guarantees for learned modular systems

Many guarantees in control depend on assumptions that are difficult to verify for high-dimensional neural networks. A key challenge is to derive guarantees for modular systems in which some components are learned, some are probabilistic, and some are hand-designed. At its core, I think this needs more emphasis on the behaviour of high dimensional latent spaces.

12.2 Measuring observability in learned representations

Information-theoretic observability criteria are attractive, but estimating mutual information in high dimensions is difficult. Practical methods require approximations, contrastive bounds, causal tests, or task-conditioned probes and knowledge of humans that may not be easy to obtain outside of controlled settings.

12.3 Excitation under human and robot constraints

Persistent excitation is easy to state but difficult to achieve safely in robotics. We need methods for collecting informative demonstrations and interventions without making the robot unsafe or the human teacher unnatural. It is also unclear to me if exciting latent spaces is sufficient to say anything about the general behaviour of the model.

12.4 Stability of distributional policies

Diffusion, energy-based, and score-based policies provide expressive action distributions, but their closed-loop stability properties are not yet well understood. A major research direction is to connect generative policy models to robust and stochastic stability theory.

12.5 The right level of modularity

Too much modularity can impose brittle assumptions and tight bottlenecks make learning painful (for the people doing the training). Too little modularity can make learning data-hungry and the resultant models inflexible. The goal is not to return to rigid pipelines, but to discover the right abstractions for each robot learning problem.

12.6 Connecting these things to practical robot problems

Robotics has a demo problem, often we construct tasks to match our solutions, rather than finding solutions that match our problems. Does any of this matter beyond the science? When we neglect the actuators, physical form of robots and the sensors they are equipped with, we are constraining ourselves to only a tiny portion of the design space.

13 Conclusion

Robot learning requires both flexibility and structure. Deep learning provides flexible function approximators and scalable amortised inference. Control theory provides a language for feedback, dynamics, observability, excitation, stability, and robustness. Probabilistic model-based machine learning provides a way to combine these ingredients as structured distributions to take a principled route to solve problems in robotics.

More generally though, this paper argues that the field must now take a further step. It is no longer enough to ask which inductive biases we should inject into robot learning systems. We must also ask which inductive biases are already present in the large learned policies that increasingly control robots. These biases are embedded in architectures, datasets, simulators, objectives, action spaces, software runtimes, and deployment constraints. They shape behaviour whether or not we understand them.

The resulting view is not classical control with neural networks attached, nor end-to-end learning with control terminology added after the fact. It is a model-based approach in which learned robot policies are treated as feedback systems with internal anatomy: circuits, manifolds, latent goals, action distributions, and stability properties that can be studied, tested, and modified.

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